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50 TestNG Interview Questions For Freshers



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1. What is TestNG?

TestNG is a testing framework inspired by JUnit and NUnit, designed for unit testing, functional testing, and integration testing. It provides powerful features such as annotations, grouping, parallel execution, and test configuration.

2. What are the key features of TestNG?

Some key features include:

- Annotations
- Test sequencing and prioritization
- Grouping tests
- Parallel test execution
- Data-driven testing
- Generating HTML reports
- Dependency testing

3. What are the main differences between TestNG and JUnit?

- TestNG provides more flexible annotations compared to JUnit.
- TestNG supports parallel testing, which is not available in JUnit.
- TestNG has built-in support for test configuration and execution sequencing.

4. What is the purpose of the @Test annotation in TestNG?

The @Test annotation marks a method as a test method. It is the primary annotation used to define test cases in TestNG.



5. What is the purpose of the @BeforeTest annotation?

@BeforeTest is used to indicate that a method should be executed before any test methods in the current <test> tag in the TestNG XML file.

6. What is the @AfterTest annotation used for?

@AfterTest indicates that the annotated method should be executed after all test methods in the current <test> tag have been executed.

7. What is the purpose of the @BeforeSuite annotation?

@BeforeSuite is used to indicate that a method should run before all the tests in the test suite.

8. What is @AfterSuite used for?

@AfterSuite is used to specify that a method should be executed after all the tests in the suite have run.

9. What are test groups in TestNG?

Test groups allow for logical grouping of test methods. You can define which group(s) a test method belongs to using the groups attribute in the @Test annotation.

10. How do you define dependencies between tests in TestNG?

You can use the dependsOnMethods or dependsOnGroups attributes in the @Test annotation to define dependencies between tests. ps allow for logical grouping of test methods. You can define which group(s) a test method belongs to using the groups attribute in the @Test annotation.



11. What is parallel execution in TestNG?

Parallel execution allows multiple test methods or classes to run simultaneously. This is configured in the TestNG XML file or through annotations.

12. How can you run tests in parallel in TestNG?

You can enable parallel execution by setting the parallel attribute in the TestNG XML file to methods, tests, or classes.

13. What is data-driven testing in TestNG?

Data-driven testing involves running a test method multiple times with different sets of input data. TestNG supports this using the `@DataProvider` annotation

14. What is the purpose of @DataProvider in TestNG?

`@DataProvider` allows a method to provide data to a test method. This enables parameterized testing where the same test runs with different input values.

15. How do you configure a test to use a DataProvider?

In the test method, you specify the `dataProvider` attribute to link it with the appropriate data provider method.

16. What is a TestNG XML file, and what is it used for?

The TestNG XML file is a configuration file used to define which tests to run, their grouping, order, parallel execution, etc.

17. What is the @BeforeMethod annotation in TestNG?

`@BeforeMethod` runs before every individual test method in a class.



18. What is @AfterMethod used for in TestNG?

@AfterMethod executes after each test method completes, typically used for cleanup operations.

19. Can TestNG generate HTML reports?

Yes, TestNG automatically generates HTML reports of the test execution, including passed, failed, and skipped tests.

20. How do you ignore a test in TestNG?

You can ignore a test by setting the enabled attribute of the @Test annotation to false.

21. What is the @BeforeClass annotation used for in TestNG?

@BeforeClass indicates that the annotated method should be run before the first test method in the current class.

22. What is @AfterClass in TestNG?

@AfterClass ensures the annotated method runs after all the test methods in the current class have executed.

23. What is the @Factory annotation used for in TestNG?

@Factory is used to create instances of test classes dynamically.

24. What is soft assertion in TestNG?

Soft assertions allow the test to continue after an assertion failure, while hard assertions halt the test immediately on failure.

25. What is a listener in TestNG?

Listeners are used to perform actions at specific points in the test execution lifecycle, such as logging or custom report generation



26. What is the IReporter interface used for in TestNG?

The IReporter interface is used to generate custom reports in TestNG.

27. How can you retry a failed test in TestNG?

You can implement the IRetryAnalyzer interface and use the retryAnalyzer attribute in the @Test annotation to retry failed tests.

28. What is the ITestResult interface in TestNG?

ITestResult is an interface used to capture the result of a test execution, including its status (pass/fail).

29. What is the use of the @Listeners annotation?

@Listeners is used to specify listener classes for a particular test class or suite.

30. What is the priority attribute in the @Test annotation?

Priority defines the order in which test methods are executed. Lower priority values are executed first.

31. How can you skip a test in TestNG?

You can throw the SkipException in a test method to skip the test.

32. What is the purpose of the @Parameters annotation in TestNG?

@Parameters is used to pass parameters from the TestNG XML file into test methods.



33. How do you pass parameters through the TestNG XML file?

You can use the `<parameter>` tag within the TestNG XML file and associate it with a test using the `@Parameters` annotation.

34. Can you run specific tests in TestNG?

Yes, you can run specific tests by specifying the class, group, or method in the TestNG XML file or using annotations.

35. What is an annotation transformer in TestNG?

Annotation transformers allow modifying annotations at runtime using the `IAnnotationTransformer` interface.

36. How does TestNG handle test dependencies?

TestNG supports dependencies between test methods using the `dependsOnMethods` and `dependsOnGroups` attributes.

37. How do you perform parallel execution at the method level in TestNG?

You can set `parallel="methods"` in the TestNG XML file to run methods in parallel.

38. What is the `@Test(singleThreaded=true)` annotation used for?

This forces all test methods in a class to run in a single thread.

39. How do you specify test execution timeouts in TestNG?

You can use the `timeOut` attribute in the `@Test` annotation to set a maximum execution time for a test method.



40. How can you handle test exceptions in TestNG?

You can handle expected exceptions using the `expectedExceptions` attribute in the `@Test` annotation.

41. How do you create a dependent test group in TestNG?

You can use `dependsOnGroups` to specify that a test group depends on the completion of another group.

42. What is the default order of execution for TestNG annotations?

The default order is:

- `@BeforeSuite`
- `@BeforeTest`
- `@BeforeClass`
- `@BeforeMethod`
- `@Test`
- `@AfterMethod`
- `@AfterClass`
- `@AfterTest`
- `@AfterSuite` transformers allow modifying annotations at runtime using the `IAnnotationTransformer` interface.

43. What is invocationCount in TestNG?

`invocationCount` is used to run the same test method multiple times.

44. What is the successPercentage attribute in TestNG?

It specifies the percentage of successful executions needed to mark a test as passed, useful in cases of flaky tests.



45. How can you exclude test methods in TestNG XML?

You can use the `<exclude>` tag within the `<methods>` tag to exclude specific test methods.

46. What is the purpose of `IInvokedMethodListener`?

It allows actions to be performed before and after the execution of test methods.

47. How do you log messages in TestNG?

You can use `Reporter.log()` to log messages during test execution, which will appear in the TestNG reports.

48. How does TestNG support cross-browser testing?

You can pass the browser type as a parameter using the TestNG XML file and run tests on multiple browsers using `@Parameters` or `@DataProvider`.

49. How do you prioritize test execution in TestNG?

You can set the priority attribute in the `@Test` annotation, with lower numbers indicating higher priority.

50. What is the role of `ISuiteListener` in TestNG?

`ISuiteListener` is used to perform actions before and after the execution of the entire test suite.





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